

# A Technical Introduction to Fraud detection

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## Section 1

Data mining to classify, cluster, and segment the data and automatically find associations and rules in the data that may signify interesting patterns, including those related to fraud. Expert systems to encode expertise for detecting fraud in the form of rules. Pattern recognition to detect approximate classes, clusters, or patterns of suspicious behavior either automatically (unsupervised) or to match given inputs. Machine learning techniques to automatically identify characteristics of fraud. Neural nets to independently generate classification, clustering, generalization, and forecasting that can then be compared against conclusions raised in internal audits or formal financial documents such as 10-Q. Other techniques such as link analysis, Bayesian networks, decision theory, and sequence matching are also used for fraud detection. A new and novel technique called System properties approach has also been employed where ever rank data is available. Statistical analysis of research data is the most comprehensive method for determining if data fraud exists. Data fraud as defined by the Office of Research Integrity (ORI) includes fabrication, falsification and plagiarism.

## Section 2

Early data analysis techniques were oriented toward extracting quantitative and statistical data characteristics. These techniques facilitate useful data interpretations and can help to get better insights into the processes behind the data. Although the traditional data analysis techniques can indirectly lead us to knowledge, it is still created by human analysts. To go beyond, a data analysis system has to be equipped with a substantial amount of background knowledge, and be able to perform reasoning tasks involving that knowledge and the data provided. In effort to meet this goal, researchers have turned to ideas from the machine learning field. This is a natural source of ideas, since the machine learning task can be described as turning background knowledge and examples (input) into knowledge (output). If data mining results in discovering meaningful patterns, data turns into information. Information or patterns that are novel, valid and potentially useful are not merely information, but knowledge. One speaks of discovering knowledge, before hidden in the huge amount of data, but now revealed. The machine learning and artificial intelligence solutions may be classified into two categories: 'supervised' and 'unsupervised' learning. These methods seek for accounts, customers, suppliers, etc. that behave 'unusually' in order to output suspicion scores, rules or visual anomalies, depending on the method. Whether supervised or unsupervised methods are used, note that the output gives

us only an indication of fraud likelihood. No stand alone statistical analysis can assure that a particular object is a fraudulent one, but they can identify them with very high degrees of accuracy. As a result, effective collaboration between machine learning model and human analysts is vital to the success of fraud detection applications.

## Section 3

Available datasets A major limitation for the validation of existing fraud detection methods is the lack of public datasets. One of the few examples is the Credit Card Fraud Detection dataset made available by the ULB Machine Learning Group.

## Section 4

Fraud represents a significant problem for governments and businesses and specialized analysis techniques for discovering fraud using them are required. Some of these methods include knowledge discovery in databases (KDD), data mining, machine learning and statistics. They offer applicable and successful solutions in different areas of electronic fraud crimes. In general, the primary reason to use data analytics techniques is to tackle fraud since many internal control systems have serious weaknesses. For example, the currently prevailing approach employed by many law enforcement agencies to detect companies involved in potential cases of fraud consists in receiving circumstantial evidence or complaints from whistleblowers. As a result, a large number of fraud cases remain undetected and unprosecuted. In order to effectively test, detect, validate, correct error and monitor control systems against fraudulent activities, businesses entities and organizations rely on specialized data analytics techniques such as data mining, data matching, the sounds like function, regression analysis, clustering analysis, and gap analysis. Techniques used for fraud detection fall into two primary classes: statistical techniques and artificial intelligence.

## Section 5

In supervised learning, a random sub-sample of all records is taken and manually classified as either 'fraudulent' or 'non-fraudulent' (task can be decomposed on more classes to meet algorithm requirements). Relatively rare events such as fraud may need to be over sampled to get a big enough sample size. These manually classified records are then used to train a supervised machine learning algorithm. After building a model using this training data, the algorithm should be able to classify new records as either fraudulent or non-fraudulent. Supervised neural networks, fuzzy neural nets, and combinations of neural nets and rules, have been extensively explored and used for detecting fraud in mobile phone networks and financial statement fraud. Bayesian learning neural network is implemented for credit card fraud detection, telecommunications fraud, auto

claim fraud detection, and medical insurance fraud. Hybrid knowledge/statistical-based systems, where expert knowledge is integrated with statistical power, use a series of data mining techniques for the purpose of detecting cellular clone fraud. Specifically, a rule-learning program to uncover indicators of fraudulent behaviour from a large database of customer transactions is implemented. Cahill et al. (2000) design a fraud signature, based on data of fraudulent calls, to detect telecommunications fraud. For scoring a call for fraud its probability under the account signature is compared to its probability under a fraud signature. The fraud signature is updated sequentially, enabling event-driven fraud detection. Link analysis comprehends a different approach. It relates known fraudsters to other individuals, using record linkage and social network methods. This type of detection is only able to detect frauds similar to those which have occurred previously and been classified by a human. To detect a novel type of fraud may require the use of an unsupervised machine learning algorithm.

### Section 6

In contrast, unsupervised methods don't make use of labelled records. Bolton and Hand use Peer Group Analysis and Break Point Analysis applied on spending behaviour in credit card accounts. Peer Group Analysis detects individual objects that begin to behave in a way different from objects to which they had previously been similar. Another tool Bolton and Hand develop for behavioural fraud detection is Break Point Analysis. Unlike Peer Group Analysis, Break Point Analysis operates on the account level. A break point is an observation where anomalous behaviour for a particular account is detected. Both the tools are applied on spending behaviour in credit card accounts. A combination of unsupervised and supervised methods for credit card fraud detection is in Carcillo et al (2019).